

Lab Sheet 6 – MPI

IT24100956

Exercise02

```
(kali@kali)-[~/SE4060-MPI-Part1]
$ mpirun --oversubscribe -np 4 ./exercise2_sum 2>/dev/null
Calculated Sum : 50000005000000
Expected Sum   : 50000005000000
Elapsed Time   : 0.000736 seconds (with 4 processes)

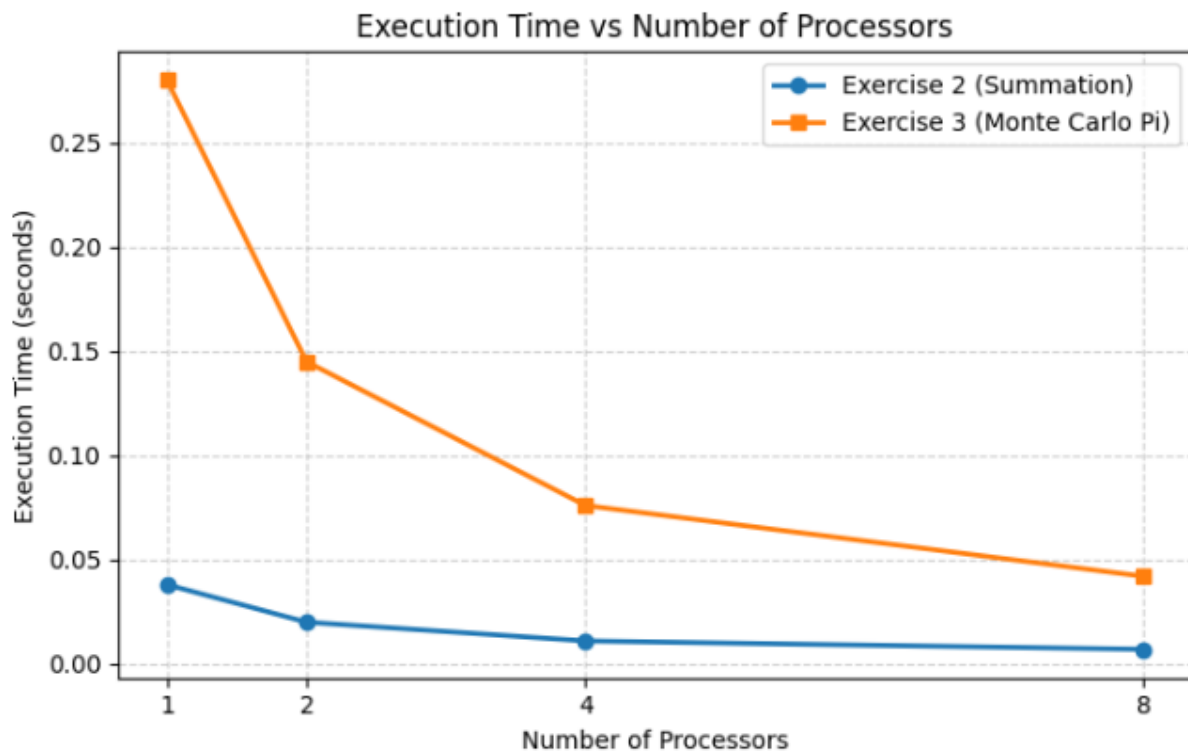
(kali@kali)-[~/SE4060-MPI-Part1]
$
```

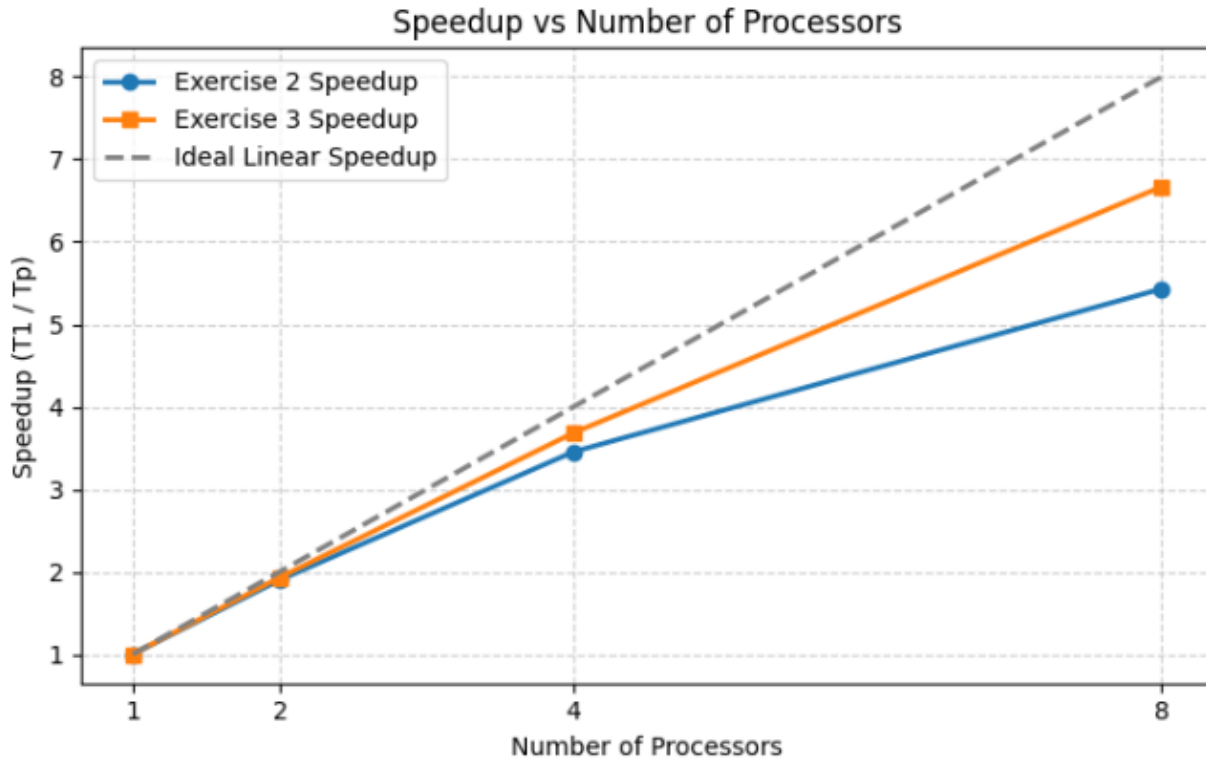
Exercise03

```
(kali@kali)-[~/SE4060-MPI-Part1]
$ mpicc -w -O3 exercise3_pi.c -o exercise3_pi 66 mpirun --oversubscribe -np 4 ./exercise3_pi 2>/dev/null
Estimated Pi : 3.140230
Elapsed Time : 0.035737 seconds (with 4 processes)

(kali@kali)-[~/SE4060-MPI-Part1]
$
```

Exercise04





Exercise05

Destination Mismatched

```
(kali@kali)~/SE4060-MPI-Part1
$ mpicc -w -O3 exercise5_mismatch.c -o exercise5_mismatch 66 mpirun --oversubscribe -np 2 ./exercise5_mismatch 2>/dev/null
[Rank 0] Sending message to Rank 1...
[Rank 1] Waiting for message from non-existent sender Rank 2...
```

Buffered Send

```
(kali@kali)~/SE4060-MPI-Part1
$ mpicc -w -O3 exercise5_bsend.c -o exercise5_bsend 66 mpirun --oversubscribe -np 2 ./exercise5_bsend 2>/dev/null
[Rank 0] Sending 12345 using MPI_Bsend
[Rank 1] Successfully received 12345
```

Exercise 6: Wildcard Receive

```
(kali@kali)-[~/SE4060-MPI-Part1]
$ mpicc -w -O3 exercise6_any_source.c -o exercise6_any_source 86 mpirun --oversubscribe -np 4 ./exercise6_any_source 2>/dev/null
[Root] Received partial count 1962816 from Rank 1
[Root] Received partial count 1964030 from Rank 3
[Root] Received partial count 1964997 from Rank 2
Estimated Pi : 3.142338
Elapsed Time : 0.041122 seconds (with 4 processes)
```

Exercise 7: Buffered Send with MPI_ANY_SOURCE

```
(kali@kali)-[~/SE4060-MPI-Part1]
$ mpicc -w -O3 exercise7_bsend.c -o exercise7_bsend 86 mpirun --oversubscribe -np 4 ./exercise7_bsend 2>/dev/null
[Root] Received 1963643 via Bsend from Rank 1
[Root] Received 1963260 via Bsend from Rank 2
[Root] Received 1964034 via Bsend from Rank 3
Estimated Pi : 3.141906
Elapsed Time : 0.031395 seconds (with 4 processes)
```